

Individual differences in crowding predict reading speed in rapid serial visual presentation

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How do individual differences in crowding relate to reading speed?

Reading speed varies more than 2-fold among typical adults. This is typically assumed to be due to language processing. Might some of it be due to visual factors?

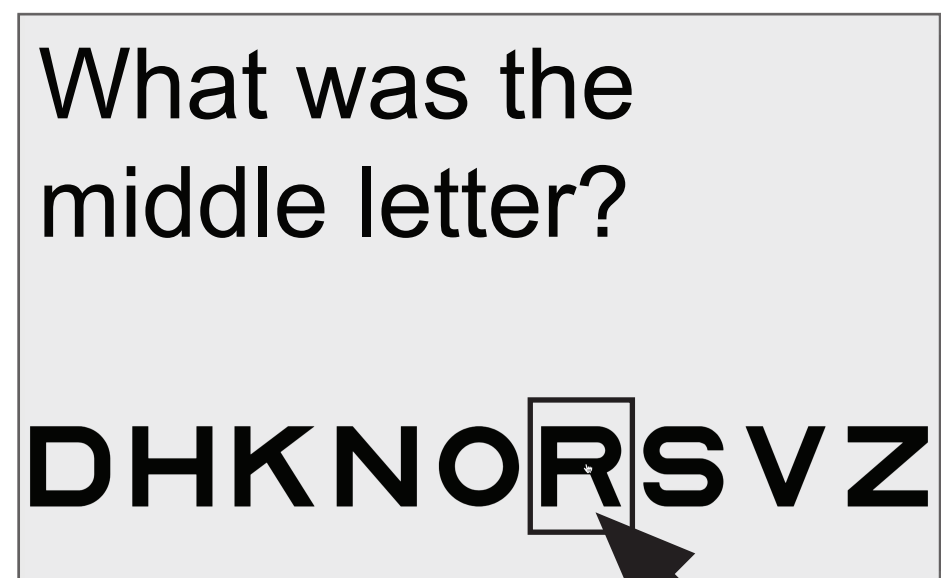
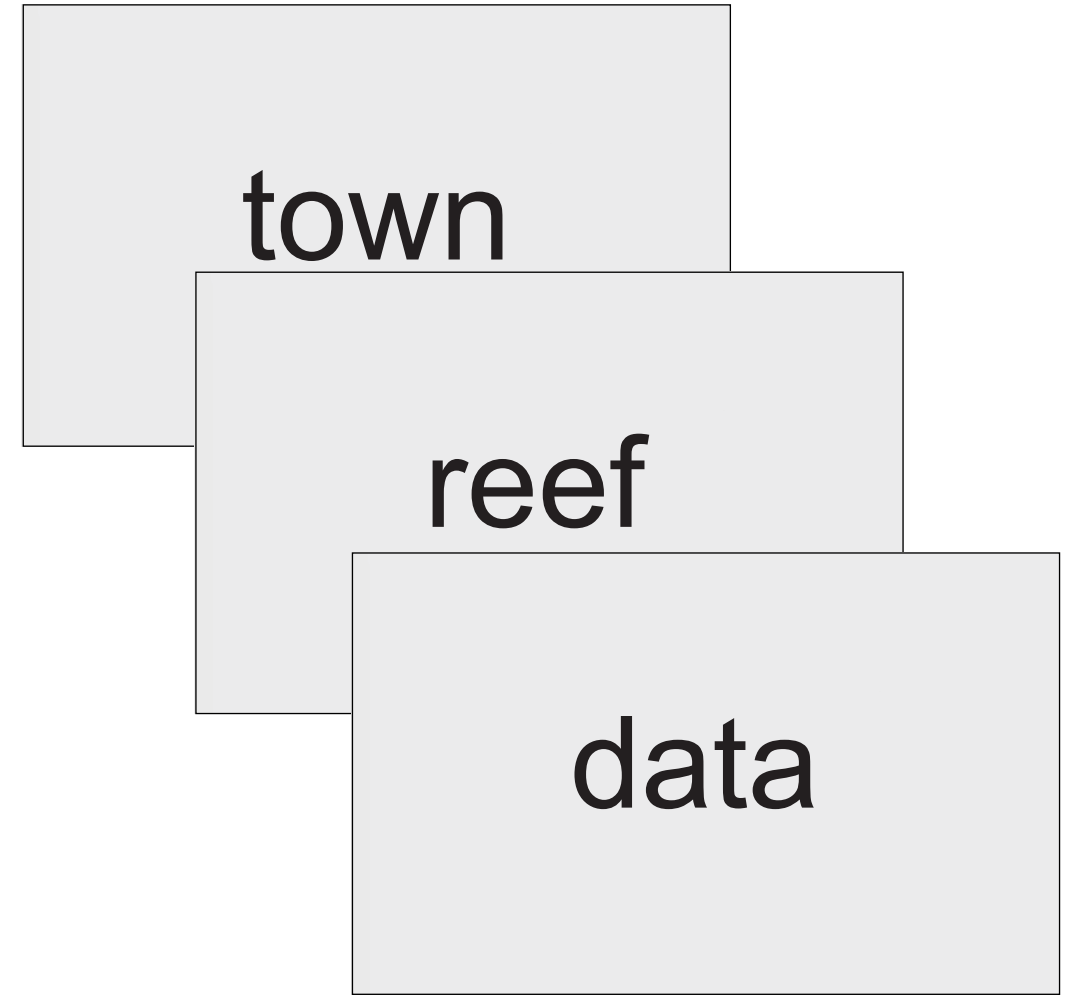
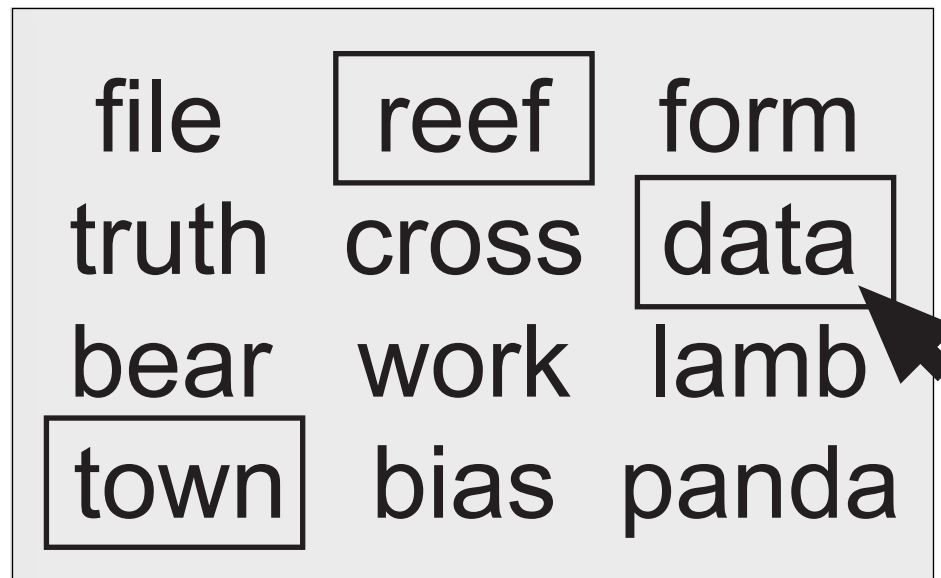
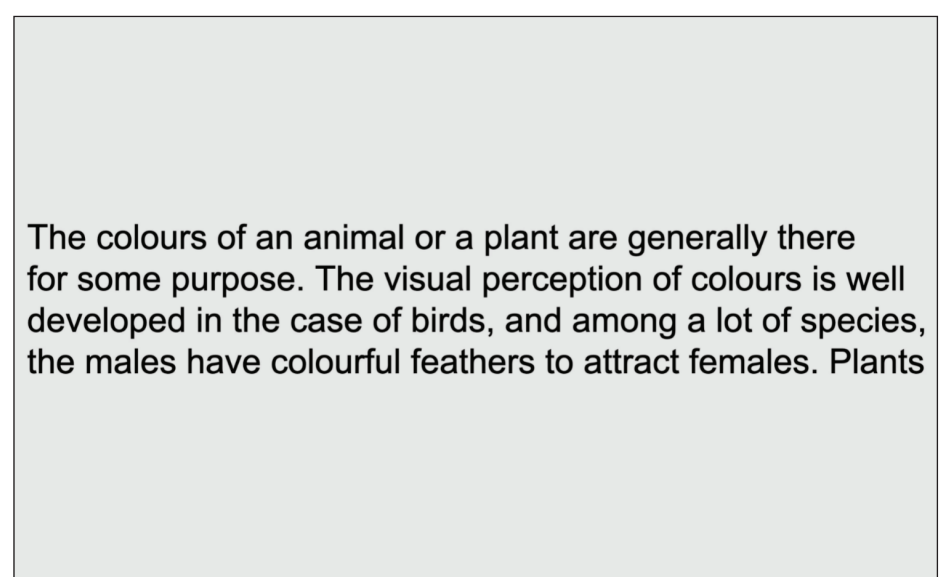
Visual crowding

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Crowding is correlated with reading speed measured with rapid serial visual presentation (RSVP):

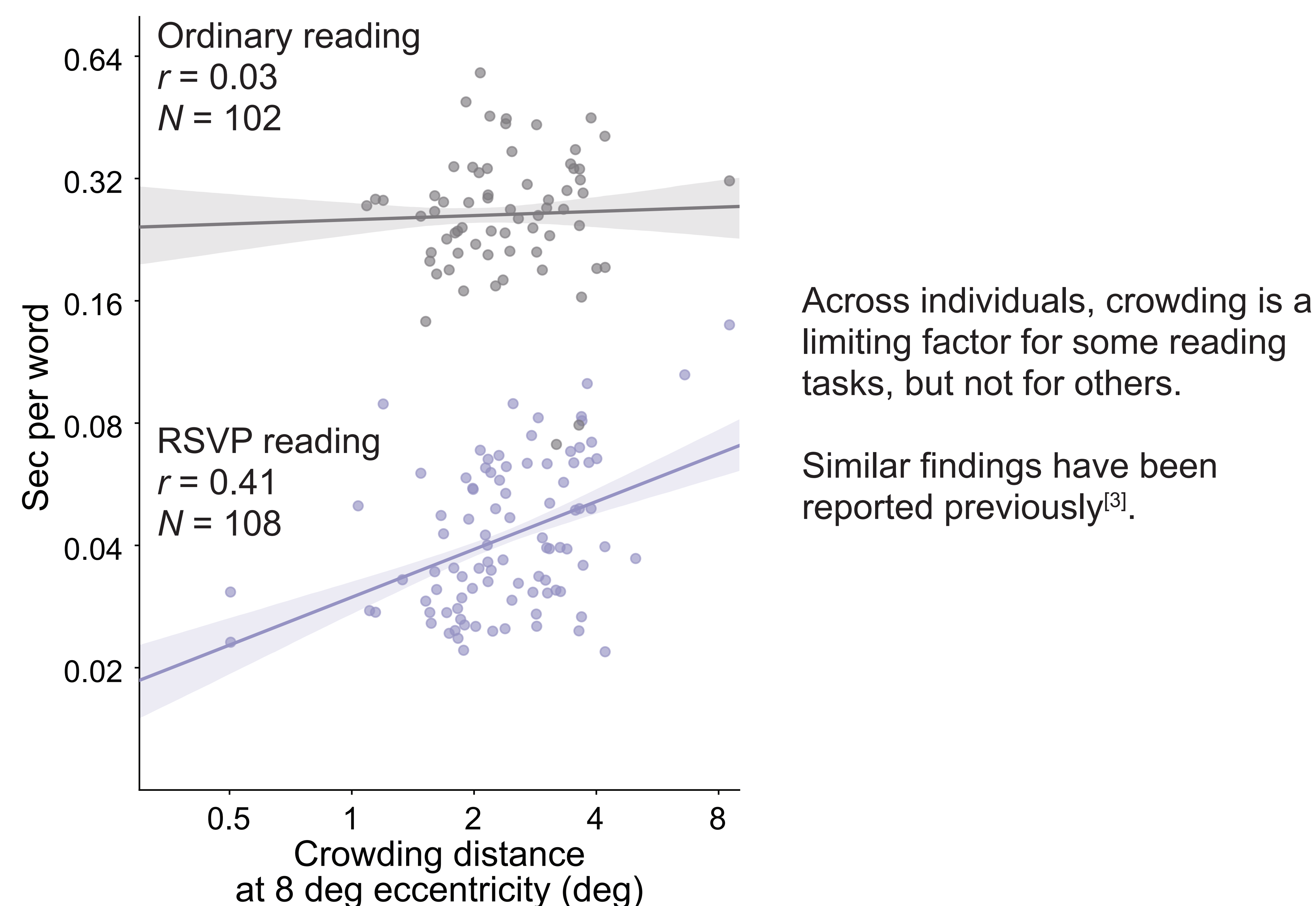
- Insufficient letter-to-letter spacing causes crowding, which greatly slows reading^[1].
- Fonts that induce higher crowding also slow reading^[2].

We measured crowding, RSVP reading, and ordinary reading in 144 adults.

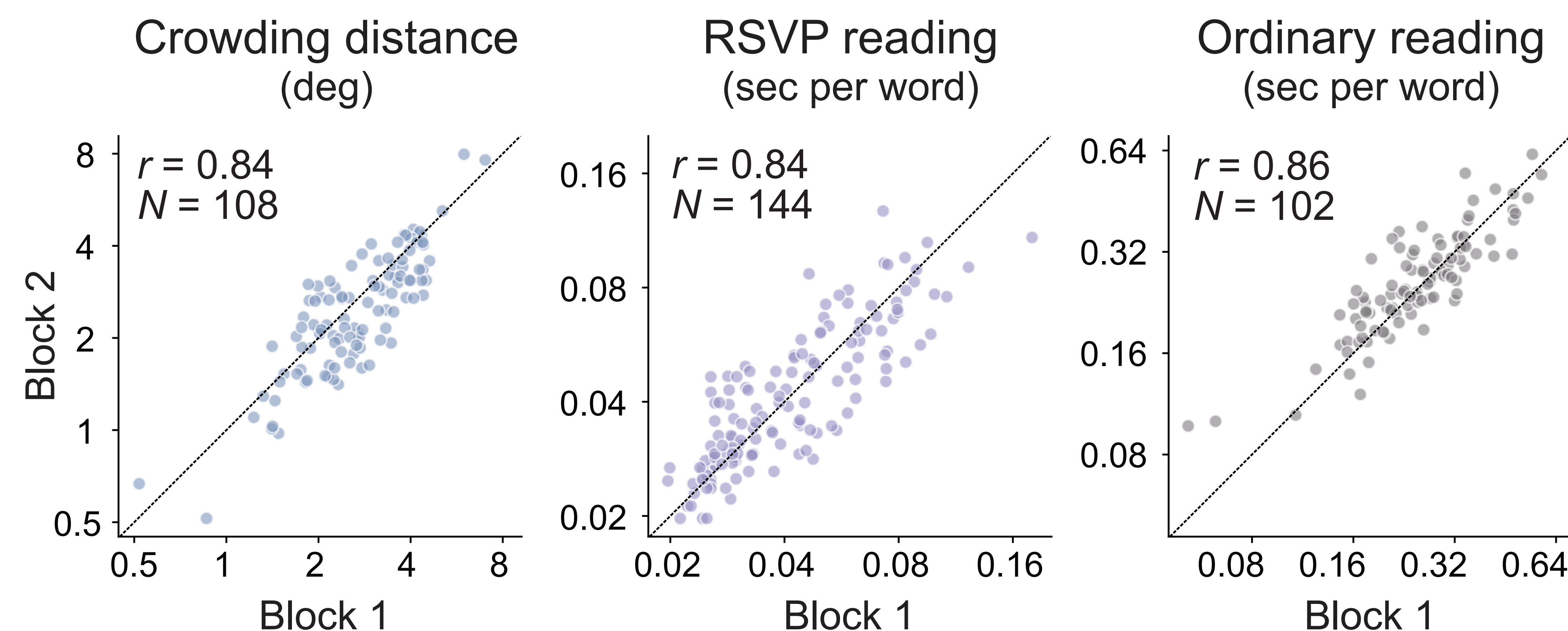
	TASK	RESPONSE	MEASURE
Crowding		What was the middle letter? 	Crowding distance (deg)
RSVP reading			Display time (sec/word)
Ordinary reading		press space	Reading time (sec/word)

The experiment was generated using the EasyEyes open-source platform for online psychophysics (<https://easyeyes.app>).

Across individuals, crowding is more correlated with RSVP than with ordinary reading.



Test-retest reliability is high ($r > 0.8$).



References: ¹Pelli, Tillman, Freeman, Su, Berger, & Majaj, 2007, *JOV*
²Bruns, Majaj, Pombo, Kim, & Pelli, 2026, *Poster(23.404)*, VSS2026
³Frömer, Dimigen, Niefind, Krause, Kliegl, & Sommer, 2015, *PLoS One*

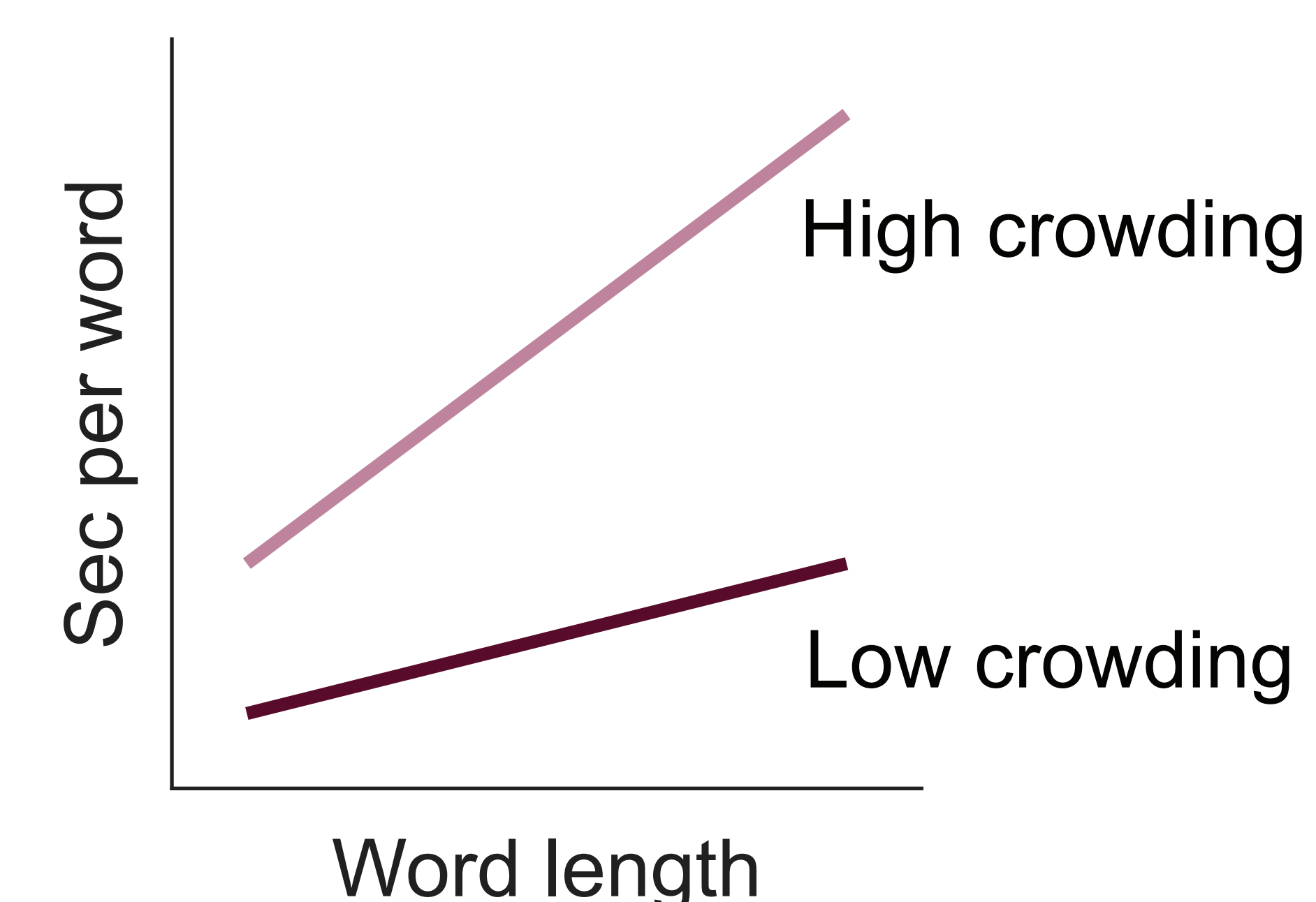
Grant support: Meta Reality Labs.

Check out related posters by Bruns (23.404) and Pelli (23.405)

Discussion: Why is crowding correlated with RSVP reading?

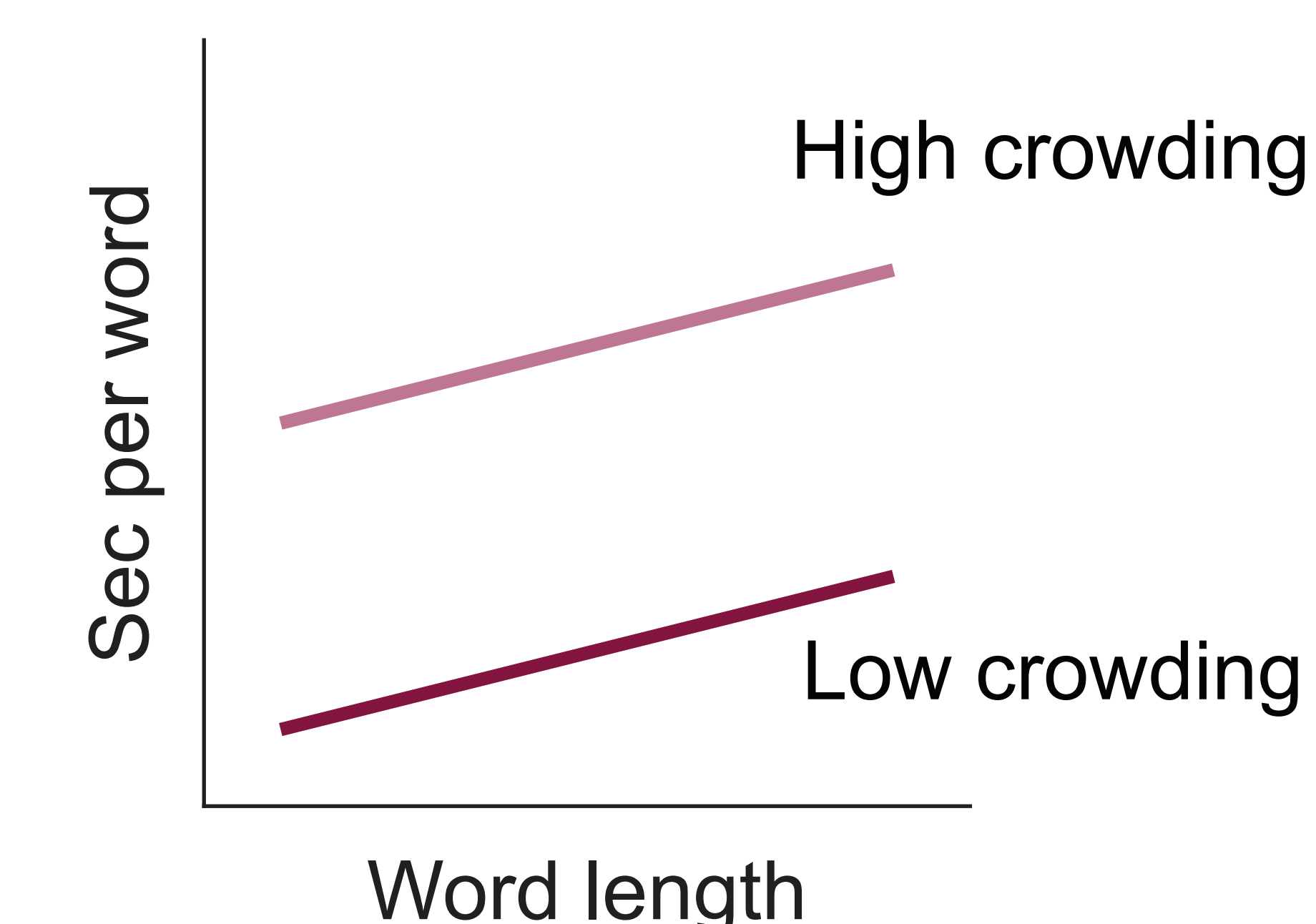
Spatial hypothesis

Crowding reduces visual span, so longer words require more fixations

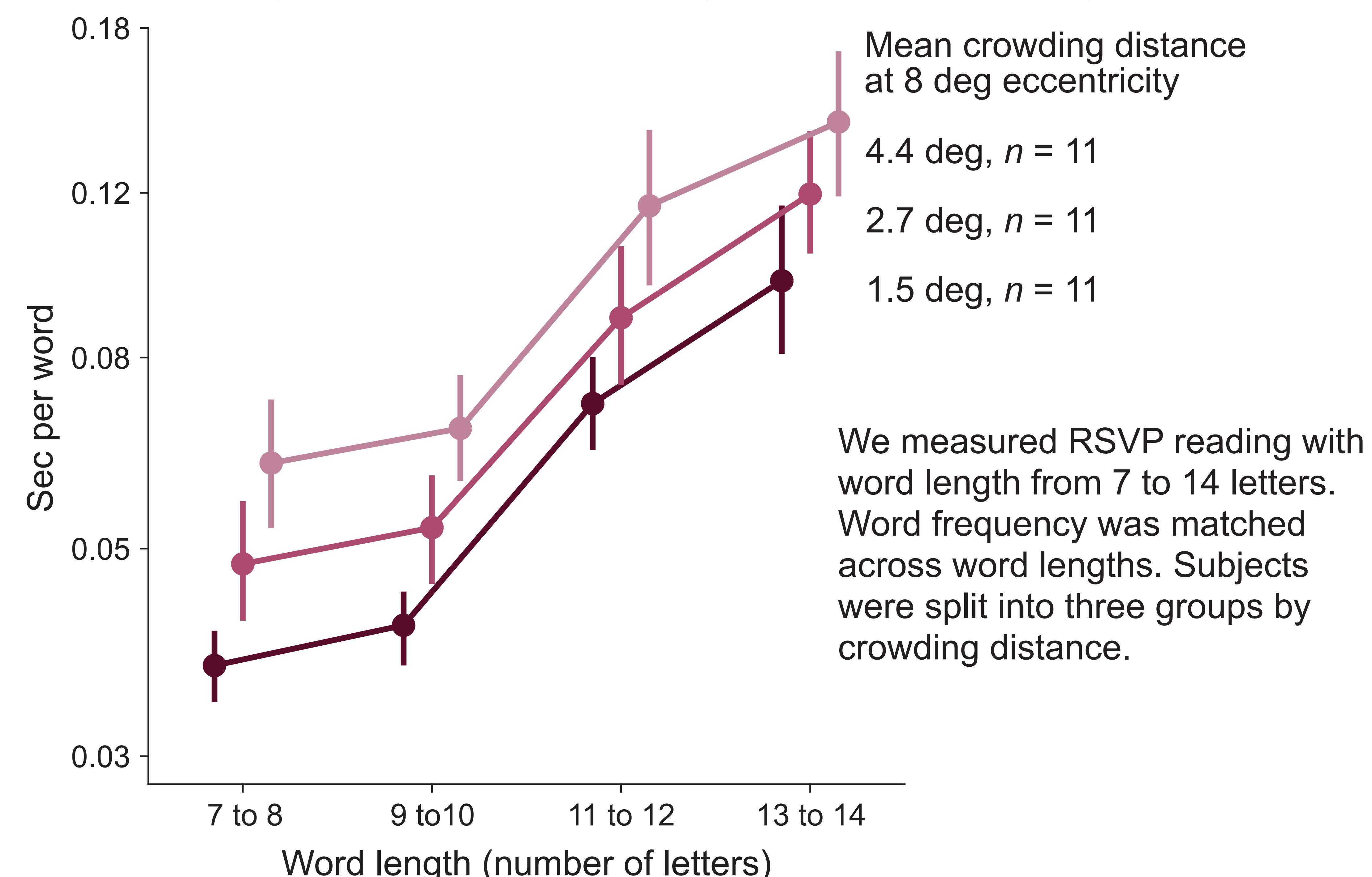


Temporal hypothesis

Crowding slows accrual, independent of word length



Preliminary data support only the temporal hypothesis.



Interpretation:

Crowding limits RSVP reading by reducing encoding efficiency.